

LoRA Fine-Tuning Enables Controllable Mathematical Reasoning in Small Language Models: A Gating Pipeline on Qwen3.5-0.8B

HanQing

Abstract—We design a gating pipeline for LoRA fine-tuning of Qwen3.5-0.8B, a 24-layer hybrid of GatedDeltaNet and Full Attention, executed entirely on CPU. The pipeline runs three ablations: layer ablation selects full-module joint fine-tuning over single-module configurations; data curriculum: AllOnce matches easy-first and is selected by the simplicity rule; LoRA rank ablation shows rank 8 matches rank 16 at small scale and outperforms it at larger scale. The gated model preserves reasoning accuracy (93% retention on int500, 99% on the showcase set) while transforming output controllability on int500: boxed usage rises from 0% to 61.2%, automatic accuracy from 0% to 12.2%, average length falls from 985 to 265 tokens (−73%), and truncation drops from 12.0% to 0%.

Index Terms—LoRA fine-tuning, Qwen3.5, mathematical reasoning, GatedDeltaNet, curriculum gating, CPU training

I. INTRODUCTION

Large language models achieve strong mathematical reasoning at scale: 7B-class systems report 64.2% on GSM8K [1], but their size precludes edge-CPU deployment. Models of 0.8B–2B fit low-resource hardware yet lag on reasoning. Qwen3.5-0.8B adopts a 24-layer hybrid attention architecture, 18 GatedDeltaNet (GDN) [2] linear-attention layers interleaved with 6 Full Attention (FA) layers. This raises three broad questions, operationalized as five research questions (RQ1–RQ5): which layer families contribute most to mathematical reasoning, how data curriculum shapes capability acquisition, and how LoRA rank and target modules interact.

We run all experiments on a commodity laptop CPU (AMD Ryzen 7 8745H, 16 GB DDR5, no GPU), with single runs costing hours to about a day and a half. This constraint makes exhaustive grid search infeasible; each run must yield one decision, which is then validated before committing compute to the next. The gating pipeline in Section II addresses this requirement.

LoRA [3] parameterizes weight updates as $W = W_0 + (\alpha/r)BA$ with $r \ll d$, cutting trainable parameters from d^2 to $2dr$. Chain-of-thought prompting [4] pushes large models further, but the small-model CPU regime remains unexplored. We fine-tune Qwen3.5-0.8B [5] on CPU with LlamaFactory 0.9.5 [6] and address them: the relative contributions of FA, GDN, and MLP modules (RQ1); easy-first versus AllOnce curricula (RQ2); rank 8 versus rank 16 at 1500 samples/1 epoch (RQ3) and 2000 samples (RQ4); and LoRA effectiveness across format regularity, output efficiency, truncation elimination, and accuracy (RQ5).

Our contributions are threefold. (1) We compare full-module joint fine-tuning against single-module fine-tuning on the hybrid architecture, and show that synchronized updates across FA, GDN, and MLP outperform any single-module configuration. (2) We design and validate the M0–M6 gating pipeline with every stage trained and evaluated on CPU. (3) We propose a four-dimensional evaluation framework and show that LoRA’s value in output controllability (format, efficiency, truncation) matches its value in accuracy.

II. THE GATING PIPELINE

Two principles govern the pipeline. First, *compute accounting*: every stage budgets its CPU hours (Table IV) and commits them only when the preceding gate has returned a validated winner, so a failed hypothesis costs one run rather than a full grid. Second, *hypothesis isolation*: each stage varies at most one factor of interest, whether LoRA target (M1), curriculum (Gate1), rank (Gate2 and M5), or scale (M5); the α/r ratio is held constant at 2 across rank comparisons, and M5’s budget difference is disclosed in Section II-E. Gating decisions follow a dominance rule: when candidates are statistically indistinguishable (Gate1: +1.2pp, $p \approx 0.46$; Gate2: identical accuracy), the simpler configuration wins: AllOnce at zero scheduling overhead, rank 8 at half the adapter parameters. Total expenditure across all runs is approximately 159 CPU hours. Fig. 1 shows the complete M0–M6 flow; only the winning configuration propagates across stages: Gate1 initializes from A’s merged weights, while Gate2 and M5 retrain from the base model.

A. M0: Data Standardization

NVIDIA Nemotron-Math [7] provides the training data: a multi-mode mathematical reasoning corpus from AoPS and StackExchange. M0 standardizes every sample to ShareGPT format, strips system prompts and emoji, and performs 100% integrity checks. M1 uses the full multi-source split; later stages use the AoPS split for distributional purity.

B. M1: Layer Ablation

Four groups share $r=8$, $\alpha=16$, $\text{lr}=10^{-4}$, $\text{epochs}=3$ on 171 unique problems (1000 samples, 5.85 answers/problem), differing only in the LoRA target: A (all linear layers), B (FA only), C (GDN only), G (MLP only). Group A achieves the highest accuracy on int500, surpassing E0 and every single-module group (Table V). Full-module joint fine-tuning gates into the data curriculum stage.

M0--M6 Gating Pipeline for Systematic LoRA Fine-Tuning

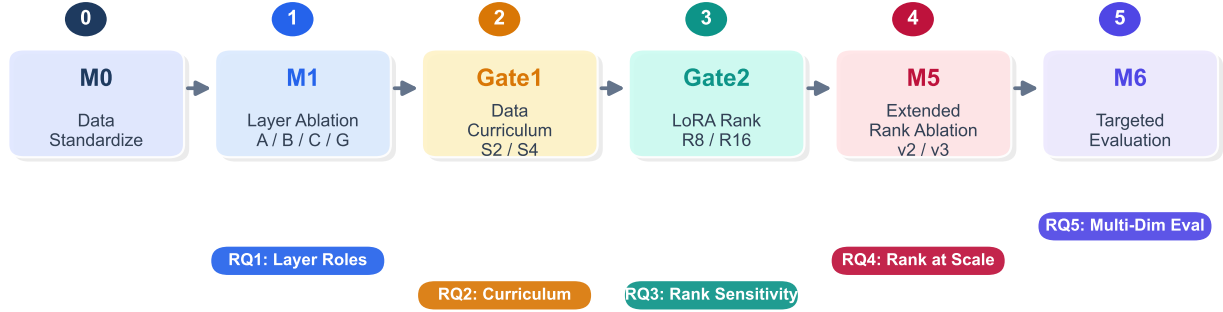


Fig. 1. M0–M6 gating pipeline. The winning configuration propagates: Gate1 initializes from A’s merged weights; Gate2 and M5 retrain from base.

C. Gate1: Data Curriculum Gate

From Group A’s merged checkpoint we compare S2 (easy-first curriculum: short-answer samples first, complex reasoning later) against S4 (AllOnce: all 1048 unique problems, 1500 samples, injected simultaneously). S4 and S2 are statistically indistinguishable (+1.2pp, $p \approx 0.46$); both eliminate truncation. The winner propagates as a data recipe, not as weights; the drop from Group A reflects the distribution shift from 171 to 1048 problems. AllOnce gates forward: a single mixed strategy suffices at this scale, with zero scheduling overhead.

D. Gate2: Rank Search Gate

R8 (rank= 8, $\alpha=16$) and R16 (rank= 16, $\alpha=32$) train on the same S4 data from base (1 epoch). Automatic accuracy is identical; R16’s boxed-usage gain is not significant. R8 suffices at half the adapter parameters and gates into M5.

E. M5: Extended Rank Ablation

M5 trains on 2000 pure Nemotron-AoPS samples (1487 unique problems, 3 epochs) from base: M5-v2 (rank 16, stopped at 2.7 epochs) versus M5-v3 (rank 8, complete 3 epochs). At 2.7 epochs (1350/1500 steps, 75 warmup), the cosine schedule yields $\text{lr} \approx 2.7 \times 10^{-6}$ (2.7% of peak), so incomplete training negligibly affects the comparison. M5-v3 outperforms M5-v2 in automatic accuracy ($p = 0.038$) and precision inside boxed outputs. This corroborates the Gate2 rank-8 decision at larger scale. Rank 8 gates into the final evaluation; ckpt-900, the M5-v3 checkpoint with the highest int500 manual accuracy, is the gated model.

F. M6: Targeted Evaluation

M6 designs a 120-problem showcase set (6 categories $\times$ 20 problems) targeting Group A’s failure modes on int500, comparing E0, A, and ckpt-900.

III. EXPERIMENTAL SETUP

A. Data and Architecture

Table I summarizes the Qwen3.5-0.8B architecture; FA layers sit at indices $\{3, 7, 11, 15, 19, 23\}$, interleaved with the 18 GDN layers so that precise retrieval and compressed memory alternate along depth.

Table II summarizes the training data. The M1 stage uses depth-oriented data (171 problems $\times$ 5.85 answers, 1000 samples); the M5 stage uses breadth-oriented data (1487 unique problems, 2000 samples). Test sets are int500 (500 GSM8K [8] problems drawn without replacement under seed 42) and the 120-problem showcase.

B. Training Configuration

Table III lists all ten runs and Table IV the CPU training cost. All models train with LlamaFactory 0.9.5 in bf16 precision: learning rate 10^{-4} , dropout 0.05, batch size 1×4 , seed 42, no packing, and single-threaded data loading. The 4096-token cutoff is derived from the tokenized maximum (max 2918 tokens, +40% margin) and covers 100% of samples. Every run starts from the same base checkpoint under seed 42, except S2/S4 which continue from Group A’s merged checkpoint (Table III); differences between runs within each gate are attributable to the gated factor alone.

TABLE I
QWEN3.5-0.8B ARCHITECTURE.

Parameter	Value
Hidden dimension d	1024
Layers L	24 (18 GDN + 6 FA)
Attention heads	8 (FA) / 16 (GDN)
partial_rotary_factor	0.25
mrope_section	[11, 11, 10]
Vocabulary	248,320
Parameters	$\approx 0.8\text{B}$

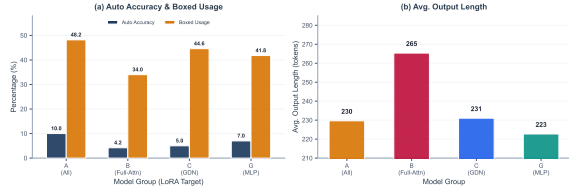


Fig. 2. M1 layer ablation: (a) automatic accuracy and boxed usage; (b) average output length. Group A leads in accuracy and boxed usage.

S2/S4 use a Double-LoRA procedure: Group A’s adapter is merged into W_0 and a second adapter trains on the merged weights, a merge-and-unload step required by Qwen3.5’s custom linear layers. Inference uses merged models, dual EOS tokens, a 4096-token generation cap, and temperature 1.0.

C. Evaluation Protocol

Evaluation uses dual-track scoring. Automatic accuracy extracts the answer the model enclosed in `\boxed{...}` delimiters; outputs without `\boxed{...}` are scored as incorrect. Manual accuracy applies a fixed four-step priority rule: (1) extract from `\boxed{...}` if present, (2) otherwise extract the last bold number, (3) otherwise match the “answer is X” pattern, (4) otherwise take the last five digits. Both metrics use deterministic string matching against the ground truth after normalization. Boxed usage rate is the proportion of outputs containing `\boxed{...}` delimiters. Automatic accuracy is the deployment-facing metric: whether the answer is automatically extractable and correct. Manual accuracy is the content-facing check. Gating decisions prioritize the deployment metric and are verified against the content metric. Significant proportion comparisons report two-sided p -values on the difference of proportions.

IV. RESULTS

A. M1: Layer Ablation

Table V and Fig. 2 report the layer ablation. Group A leads in both accuracy metrics; every single-module group underperforms the full-module configuration. All LoRA fine-tuned models eliminate truncation and compress output length by 73–77%.

B. Gating Decisions

Gate1 selects AllOnce: S4 at 7.6% exceeds S2 at 6.4% (+1.2pp, $p \approx 0.46$); neither truncates. Gate2 selects rank 8: R8 and R16 produce identical automatic accuracy (7.4%); both gates favor the simpler configuration.

TABLE II
DATASETS AND TRAINING SCALE.

Stage	Data	Unique	Ans/Q	Samples
A/B/C/G	Nemotron (multi-source)	171	5.85	1000
S2/S4	Nemotron-AoPS	1048	1.43	1500
R8/R16	Nemotron-AoPS	1048	1.43	1500
M5-v2/v3	Nemotron-AoPS	1487	1.34	2000

TABLE III
TRAINING CONFIGURATIONS. ALL GROUPS SHARE LR= 10^{-4} , DROPOUT= 0.05, A 4096-TOKEN CUTOFF, BATCH= 1×4 , BF16.

Group	Target	r/α	Start	Epochs	Data
A	all	8/16	base	3	171×5.85
B	FA	8/16	base	3	171×5.85
C	GDN	8/16	base	3	171×5.85
G	MLP	8/16	base	3	171×5.85
S2	all	8/16	A merged	1	1048/1500
S4	all	8/16	A merged	1	1048/1500
R8	all	8/16	base	1	1048/1500
R16	all	16/32	base	1	1048/1500
M5-v2	all	16/32	base	2.7	1487/2000
M5-v3	all	8/16	base	3	1487/2000

C. M5: Rank at Scale

The rank comparison reverses the small-scale tie. M5-v3 (rank 8) reaches 13.8% automatic accuracy versus 9.6% for M5-v2 (rank 16), a +4.2pp gain ($p = 0.038$). Rank 16’s extra capacity leads to longer outputs (319.95 vs. 270.21 tokens) and higher boxed usage (68.8% vs. 62.2%), yet its precision inside boxed outputs is 13.9% versus 22.2%: longer generations contain more arithmetic errors. Rank 8 gates into the final evaluation; ckpt-900, the M5-v3 checkpoint with the highest int500 manual accuracy at 34.8%, is the gated model.

D. Showcase Evaluation

Table VI reports the 120-problem comparison. ckpt-900 achieves the highest manual accuracy among the fine-tuned models, recovering 99% of base-level accuracy on the targeted set, whereas A retains only 89%. Table VII disaggregates by category: ckpt-900 outperforms A in 5 of 6 categories; A leads only in variable identification.

E. Checkpoint Dynamics

M5-v3’s four checkpoints separate training from validation signals (Table VIII and Fig. 3). ckpt-900 reaches the highest manual accuracy despite only the third-lowest eval_loss; ckpt-1050 has the lowest eval_loss but lower accuracy. Train loss shows a negative trend with manual accuracy ($r = -0.88$,

TABLE IV
TRAINING TIME ON CPU (AMD RYZEN 7 8745H, 16 GB RAM).

Model	Steps	Epochs	Time (h)	s/step
A	750	3	12.86	61.7
B	750	3	12.72	61.1
C	750	3	10.84	52.0
G	750	3	12.92	62.0
S2	375	1	12.06	115.8
S4	375	1	10.46	100.4
R8	375	1	12.44	119.4
R16	375	1	11.89	114.2
M5-v2	1350	2.7	≈ 30	≈ 80
M5-v3	1500	3	33.22	79.7

TABLE V
M1 LAYER ABLATION ON INT500 (500 SAMPLES).

Group	Auto	Manual	Boxed	Trunc	Avg tokens
E0	0.0%	37.4%	0.0%	60/500	984.65
A	10.0%	42.6%	48.2%	0/500	229.64
B	4.2%	30.2%	34.0%	0/500	265.35
C	5.0%	34.0%	44.6%	0/500	231.03
G	7.0%	37.2%	41.8%	0/500	222.70

TABLE VI
SHOWCASE SET COMPARISON (120 PROBLEMS).

Model	Manual	Boxed	Trunc	Avg tokens
E0	82.5%	0.0%	1	306.84
A	73.3%	42.5%	0	137.31
ckpt-900	81.7%	55.8%	0	153.98

$n = 4$), though the relationship is not monotonic at the checkpoint level. Eval_loss is not a reliable proxy for downstream performance.

V. DISCUSSION

Synchronization hypothesis. Single-module updates break the hybrid architecture’s balance: FA, GDN, and MLP evolve against stale representations of the other two. Group A’s lead of 42.6% vs. 30.2–37.2% is consistent with synchronized joint evolution.

Accuracy–efficiency trade-off. Fine-tuning affects more than accuracy: boxed usage rises on both test sets, average length drops from 984.65 to 264.88 tokens (−73.1%) on int500 and from 306.84 to 153.98 (−49.8%) on the showcase, and truncation is eliminated. At 27% of the base token budget, the gated model retains 93% of manual accuracy on int500. On the showcase set, it recovers base-level accuracy (81.7% vs. 82.5%) that A lost (73.3%). Group A saw 171 problems at 5.85 answers each (depth-oriented); M5 saw 1487 at 1.34 each (breadth-oriented). The pipeline trades peak int500 performance for category-level robustness.

VI. CONCLUSION

We present an M0–M6 gating pipeline for LoRA fine-tuning of Qwen3.5-0.8B on CPU. Full-module joint fine-tuning reaches 42.6% manual accuracy (+5.2pp over base), outperforming all single-module configurations. Rank 8 matches rank

TABLE VII
PER-CATEGORY ACCURACY (20 PROBLEMS PER CATEGORY).

Category	E0	A	ckpt-900	ckpt-900–A
Unit ID	15/20	12/20	16/20	+20pp
Multi-factor mult.	17/20	12/20	14/20	+10pp
Variable ID	16/20	17/20	16/20	−5pp
Multi-step complete	15/20	13/20	15/20	+10pp
Rate/ratio	18/20	18/20	19/20	+5pp
Arithmetic	18/20	16/20	18/20	+10pp

TABLE VIII
CHECKPOINT DYNAMICS: EVAL_LOSS DOES NOT TRACK ACCURACY.

Ckpt	Tr.	Ev.	Man.	Auto	Box	Tok.
ckpt-450	0.5551	0.5530	26.0%	7.0%	58.6%	310.06
ckpt-900	0.4659	0.5535	34.8%	12.2%	61.2%	264.88
ckpt-1050	0.4694	0.5519	34.6%	12.4%	62.0%	275.54
ckpt-1500	0.4297	0.5606	33.4%	13.8%	62.2%	270.21

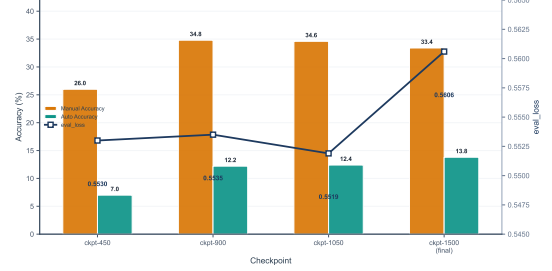


Fig. 3. Checkpoint metrics: eval_loss bottoms out at ckpt-1050 while manual accuracy peaks at ckpt-900.

16 at small scale and surpasses it at larger scale (13.8% vs. 9.6%, $p = 0.038$). The gated model preserves reasoning accuracy (93% on int500, 99% on showcase) while transforming controllability on int500: boxed usage 0%→61.2%, automatic accuracy 0%→12.2%, length −73%, truncation 12.0%→0%.

REFERENCES

- [1] Z. Shao, P. Wang, Q. Zhu *et al.*, “Deepseekmath: Pushing the limits of mathematical reasoning in open language models,” arXiv:2402.03300, 2024.
- [2] S. Yang, J. Kautz, and A. Hatamizadeh, “Gated delta networks: Improving mamba2 with delta rule,” arXiv:2412.06464, 2024.
- [3] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, and W. Chen, “Lora: Low-rank adaptation of large language models,” in *Proc. Int. Conf. on Learning Representations (ICLR)*, 2022.
- [4] J. Wei, X. Wang, D. Schuurmans *et al.*, “Chain-of-thought prompting elicits reasoning in large language models,” in *Proc. Advances in Neural Information Processing Systems (NeurIPS)*, 2022.
- [5] Q. Team, “Qwen3 technical report,” arXiv:2505.09388, 2025.
- [6] Y. Zheng, R. Zhang, J. Zhang *et al.*, “Llamafactory: Unified efficient fine-tuning of 100+ language models,” in *Proc. Annual Meeting of the Association for Computational Linguistics (ACL)*, 2024.
- [7] W. Du, S. Toshniwal, B. Kisanin, S. Mahdavi, I. Moshkov *et al.*, “Nemotron-math: Efficient long-context distillation of mathematical reasoning from multi-mode supervision,” arXiv:2512.15489, 2025.
- [8] K. Cobbe, V. Kosaraju, M. Bavarian, M. Chen, H. Jun *et al.*, “Training verifiers to solve math word problems,” arXiv:2110.14168, 2021.